

52) Show that if A, B and C are sets, then
 $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

Solution:

$$\begin{aligned}
 |A \cup B \cup C| &= |A \cup B| + |C| - |(A \cup B) \cap C| \\
 &= |A| + |B| - |A \cap B| + |C| - |(A \cup B) \cap C| \quad (2^{\text{nd}} \text{ Distributive Law}) \\
 &= |A| + |B| - |A \cap B| + |C| - |(A \cap C) \cup (B \cap C)| \\
 &= |A| + |B| - |A \cap B| + |C| - |A \cap C| - |B \cap C| + |(A \cap C) \cap (B \cap C)| \\
 &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |(A \cap C) \cap B| \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |(B \cap (A \cap C)) \cap C| \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |(B \cap A) \cap C| \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |(B \cap A) \cap (C \cap C)| \quad (2^{\text{nd}} \text{ Associative Law}) \\
 &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |(A \cap B) \cap (C \cap C)| \quad (2^{\text{nd}} \text{ Commutative Law}) \\
 &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |(A \cap B) \cap C| \quad (2^{\text{nd}} \text{ Idempotent Law}) \\
 &= |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C| \quad \square
 \end{aligned}$$

(This is a special case of the inclusion-exclusion principle, which will be studied in Chapter 7).

53) Let $A_i = \{1, 2, 3, \dots, i\}$ for $i = 1, 2, 3, \dots$. Find
 a) $\bigcup_{i=1}^n A_i$ b) $\bigcap_{i=1}^n A_i$.

Solution:

$$\begin{aligned}
 A_1 &= \{1\} & A_2 &= \{1, 2\} & \dots & A_i &= \{1, 2, 3, \dots, i\} & \dots & A_n &= \{1, 2, 3, \dots, n\} \\
 \bigcup_{i=1}^n A_i &= A_1 \cup A_2 \cup \dots \cup A_n = \{1, 2, 3, \dots, n\}
 \end{aligned}$$

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap \dots \cap A_n = \{1\} \quad \square$$

57) Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i ,

a) $A_i = \{-i, -i+1, \dots, -1, 0, 1, \dots, i-1, i\}$
 $A_1 = \{-1, 0, 1\}$ $A_2 = \{-2, -1, 0, 1, 2\}$ $A_3 = \{-3, -2, -1, 0, 1, 2, 3\}$
 $A_i = \{-i, -i+1, \dots, -1, 0, 1, \dots, i-1, i\}$ $A_{\infty} = \{-\infty, \dots, -1, 0, 1, \dots, \infty\}$
 $\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \cup \dots = \{-\infty, \dots, -1, 0, 1, \dots, \infty\} = A_{\infty} = \mathbb{Z}$
 $\bigcap_{i=1}^{\infty} A_i = A_1 \cap A_2 \cap A_3 \cap \dots = \{-1, 0, 1\} \quad \square$

b) $A_i = \{-i, i\}$ $A_1 = \{-1, 1\}$ $A_2 = \{-2, 2\}$ $A_3 = \{-3, 3\}$ $\dots$ $A_i = \{-i, i\}$
 $A_{\infty} = \{-\infty, \infty\}$ $\bigcup_{i=1}^{\infty} A_i = A_1 \cup A_2 \cup A_3 \cup \dots = A_{\infty} = \{-\infty, \infty\}$
 $= \mathbb{Z} - \{0\}$